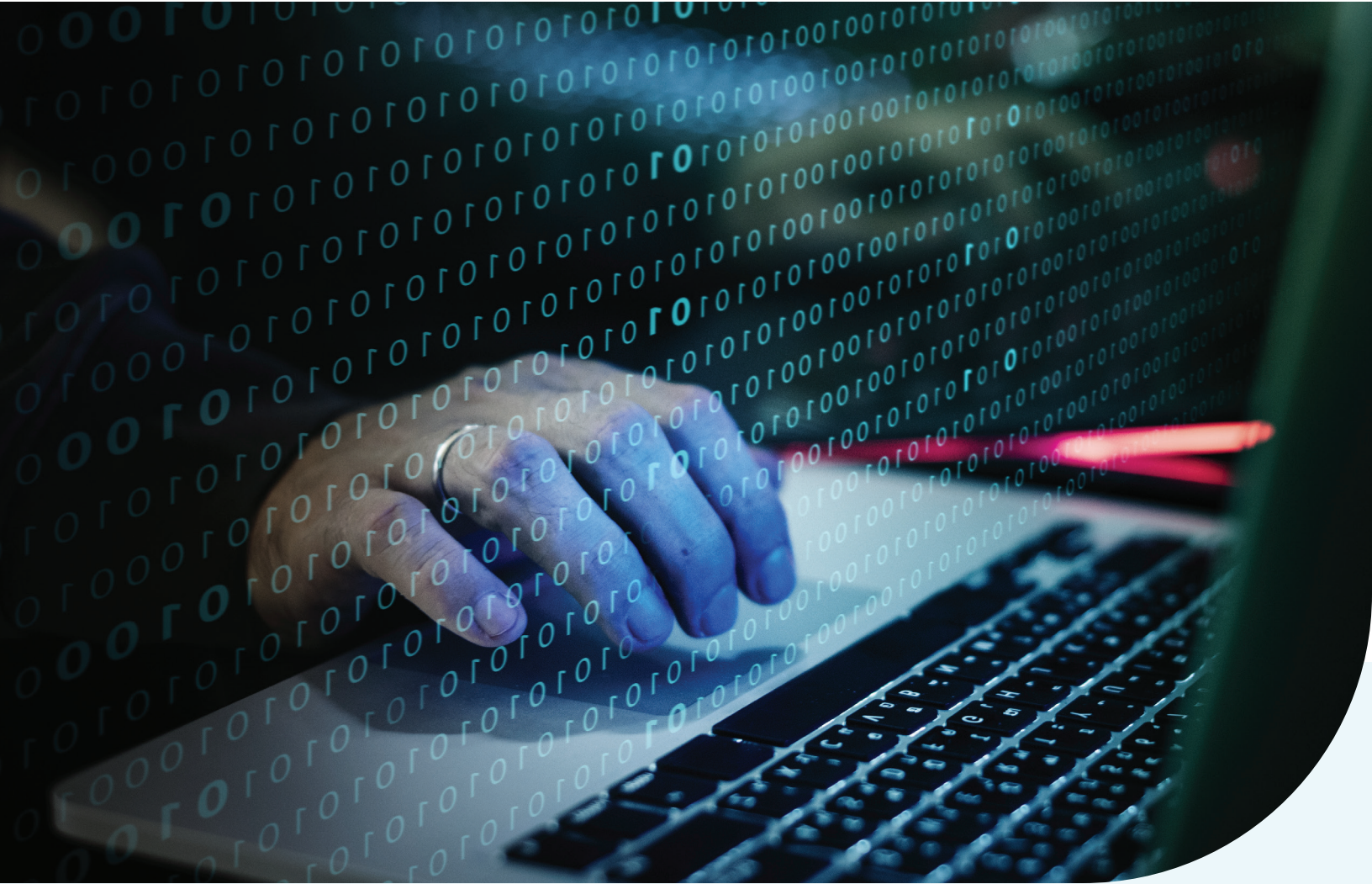




Ethical Hacking Essentials (EHE)

Lesson Plan

**EC-Council Academia**

Student Course Syllabus

Ethical Hacking Essentials

Instructor Contact Information

Instructor Name:		Office Location:	
Email:		Phone Contact:	
Office Hours:			

I. Purpose of Class:

This class is designed to provide students to establish and govern minimum standards for credentialing entry-level information security specialists in ethical hacking measures, inform the public that credentialed individuals meet or exceed the minimum standards, and reinforce ethical hacking as a unique and self-regulating profession.

II. Course Objectives:

Students who successfully complete this class will be able to:

1. Discuss the key issues plaguing the information security world and review various information security laws and regulations.
2. Comprehend Cyber Kill Chain Methodology, hacking and ethical hacking concepts, hacker classes, different phases of a hacking cycle, and assess essential ethical hacking tools.
3. Examine various information security threats and vulnerabilities, identify different types of malware and perform vulnerability assessments.
4. Explain and use different password cracking, social engineering, insider threats, and identity theft techniques, and discuss their countermeasures.
5. Examine various network-level attacks, including sniffing, denial-of-service, and session hijacking, and discuss their countermeasures.
6. Examine various application-level attacks, including webserver exploitation, OWASP top-10 attacks, and SQL injection, and discuss their countermeasures.
7. Discuss different types of wireless encryption, examine wireless threats and attacks, and suggest countermeasures.

8. Describe various mobile platform attack vectors, mobile device management, mobile security guidelines, and identify essential mobile security tools.
9. Discuss IoT and OT concepts, examine various IoT and OT threats and attacks, and suggest countermeasures.
10. Describe various Cloud computing technologies, examine cloud computing threats and attacks, and identify security techniques.
11. Discuss fundamentals of penetration testing, its benefits, strategies, and phases, and examine guidelines for penetration testing.

III. EHE Grading Policy:

The academic credit used by EHE is based off twelve-week courses that constitute 3 credits per course. All courses constitute approximately 100 hours of work per course (reading, Online Discussion Forum postings, assignments, papers and supplementary requirements).

Point-grade breakdown:

The EHE Course grading structure will follow the Academic Grading Symbols and Quality Points Earned.

Percentage	Points	Grade	Comments	Quality Points Earned
90 - 100%	360-400	A	Superior Attainment of Course Outcomes	4.0
80 - 89%	320-359	B	Good Attainment of Course Outcomes	3.0
70 - 79%	280-319	C	Acceptable Attainment of Course Outcomes	2.0
60 - 69%	240-279	D	Poor Attainment of Course Outcomes	1.0
0 - 59%	000-239	F	Non-Attainment of Course Outcomes	0.0

IV. EHE Grading Structure

S.No.	Assessment	Details	Points
1	Course Discussion Threads	There are 10 discussion threads (each is worth 4 pts.)	40
2	Assignments from Readings	There are 14 Readings (each question is worth 6 pts.)	84
3	Lab Assignments	There are 23 lab assignments (each assignment is worth 4 pts.)	92
4	Course Quizzes	There are 34 quiz questions (each question is worth 1 pt.)	34
5	Research Project	There is 1 Research Project (worth 50 pts.)	50
6	Final Exam	There is 1 Final Exam (worth 100 pts.)	100
Total Points Possible:			400

NOTE: Each week you will have assessments that you must complete for class credit.

V. EHE PROGRAM OBJECTIVES:

1. Application of technical strategies, tools and techniques to secure data and information for a customer or client
2. Adherence to a high standard of ethical behavior
3. Use of research in both established venues and innovative applications to expand the body of knowledge in information assurance
4. Application of principles of critical thinking to creatively and systematically solve the problems and meet the challenges of the ever-changing environments of cyber security
5. Mastery of the skills necessary to move into leadership roles in companies, agencies, divisions, or departments

VI. LEARNING STANDARDS:

Four Standards of Performance:

Always use these standards of performance anytime you write a research paper, respond to a case study or answer discussion threads.

These 4 Standards of Performance, when used fully, will produce an excellent paper.

1. **Depth of Reflection:** Be sure to demonstrate a conscious and thorough understanding of the writing prompt and subject matter. Analyze the evidence from several perspectives, draw conclusions based on evidence, research and/or experience.

2. **Sequence your information:** Your work has a clear and appropriate beginning, development and conclusion. Most information is organized in a clear and logical way. There is a beginning and a conclusion. Paragraphing and transitions are adequate.
3. **Conclusions:** Draw conclusions based on evidence, research or experience. Use the context of your readings and the assumptions the author makes.
4. **Language Use:** Use sophisticated language that is precise and interesting. Sentence structure must be correct to the grammar of the language.

VII. ADDITIONAL NOTES AND EXPECTATIONS:

Attendance

Documenting Attendance in Distance Education Courses

Changes to federal regulations in recent years have highlighted the importance of accurate reporting on student attendance and participation in all courses, including distance education courses (online and hybrid/blended courses).

In order to comply, We provide the following procedural guidance to faculty when filling out your non-participation rosters. A student attends an online course (or the online portion of a hybrid/blended course) by participating in class or otherwise engaging in an academically related activity. Examples of such activity include but are not limited to:

1. Contributing to an online discussion or text chat session
2. Submitting an assignment or working draft; working through exercises
3. Taking a quiz or exams
4. Viewing and/or completing a tutorial
5. Initiating contact with a faculty member to ask a course-related question.

Such academically related activities are readily tracked and documented through the University learning management system, email system, and in some cases publisher websites. Documenting that a student has logged into an online class or website is not sufficient, by itself, to demonstrate academic attendance by the student. For example, if a student simply logs into an online course on September 15 and logs out, without any further activity, the student did not attend the online class on that day.

Late Assignments Late assignments are not in the student's best interest. In that regard, if you have a problem that will not allow you to complete the assignments in a timely manner, please send the instructor an immediate request and explanation. It is the instructor's option, decision, and discretion to accept or reject the student's request. Continued requests will not be accepted and the student may be dropped from the course at the instructor's request.

Writing Skills If your writing skills are not the best, there are some modules under "Student Orientation" in the LMS that can help you prepare for your writing process or you may want to purchase a basic writing skills book that will help you.

ACADEMIC HONESTY POLICY

Course Policies On Cheating and Plagiarism

As a model of the highest ethical standards and as an institution of higher learning, University expects its students to conduct themselves with an unquestionable level of honesty and integrity. University will not tolerate academic cheating or plagiarism in any form. Learning to think and work independently is not only a part of the educational process; it is the educational process. Cheating or plagiarism in any form is considered a serious violation of university policy, which each student agreed to when accepted into the program. Student academic behaviors that violate the university policy will result in disciplinary action, without exception.

Some examples of cheating and plagiarism include but are not limited to:

Cheating	Use of material, information, or study aids not permitted by the faculty
Plagiarism	Use of another's words or ideas without acknowledging the source of the information
Falsification or Fabrication	Changing or altering data, quotes, citations, grades or academic records
Unauthorized Collaboration	Intentional sharing of information when the faculty does not approve of such collaboration

University will take action in all cases of academic dishonesty. The first instance will result in a failing grade for the assignment. The second instance will result in a failing grade in the class. The third instance will result in dismissal from the university. A record of all instances of academic dishonesty for University, as well as a record of the action taken, will be kept in the individual student file and Dean's file.

1. Steps to be taken in the instance of academic dishonesty are: The faculty/staff will inform the student of the allegation and provide evidence, offering the student the opportunity to respond and/or rectify the issue depending on the nature of the dishonesty and the particular assignment.
2. Once the student has had a chance to respond, the faculty/staff will determine if academic dishonesty has occurred. If the faculty/staff concludes that academic dishonesty has occurred and has proof, they will report the student's name, the class and assignment, the nature of the academic dishonesty and the proof to the Dean. The type of disciplinary action to be taken will be determined by the student's record of instances identified above and will be applied by the faculty and/or the Dean.

Citing Sources

In academic communities, the ethics of research demand that writers be credited for their work and their writing. To not do so is to plagiarize- to intentionally or unintentionally appropriate the ideas, language, or work of another without sufficient acknowledgement that such material is not one's own. Whenever a student quotes, paraphrases, summarizes, or otherwise refers to the

work of another, the student must cite his or her source either by way of parenthetical citation or footnote. Unfortunately, this is the most common form of academic dishonesty, but regardless it will be responded to with failing grades or dismissal.

Original Content

Students are expected to create their discussion topics, assignments and essays using the majority of their own personal thoughts and ideas. All works must contain a minimum of 75% original work. Any work submitted that contains more than 25% unoriginal work regardless of whether the sources are cited properly may be considered a violation of the academic honesty policy, depending on the nature of the assignment, and consent of the assigned instructor.

Timeline

Discovery of violation of the Academic Honesty Policy can occur at any time. Issuance of a grade, or even degree, can be changed if it is discovered that an academic honesty violation occurred. *The bottom line is this; it's just not worth it.*

Student Identity Verification

- University takes measures to verify the identity of the students who are applying to the university, completing courses, and taking proctored exams.
- Students access their courses and reference materials through our secure online learning management system, where they are required to enter their username and password. Students are responsible for the safeguard of their individual credentials.
- University implements student identity verification in several ways to ensure proper ID.
 1. A Valid Government issued ID is required with admissions application.

Login credentials are required for the online LMS portal

VIII. REQUIRED TEXTS THROUGHOUT THIS COURSE:

The following texts will be required for this course:

- Ethical Hacking Essentials (EHE) v1 Academia Series

These are embedded in the General Section of your course. The corresponding chapters for each week are embedded in your weekly readings.

IX. SCHEDULED OUTLINE OF COURSE TOPICS**WEEK ONE:****MODULES COVERED:**

- Module 01 – Information Security Fundamentals

WEEK'S OBJECTIVES USED:

1. Discuss the key issues plaguing the information security world and review various information security laws and regulations..

WEEK 1 ASSESSMENTS:

These are found in the course under Week One.

- Discussion Threads: 1 discussion thread (4 pts.)
- Assignments from Readings: 3 Readings (each worth 6 pts.)
- Quizzes: 3 quiz questions (each question is worth 1 pt.)

DISCUSSION THREAD:

1. Discuss classification of attacks.

ASSIGNMENTS FROM READINGS:

1. Discuss motives, goals and objectives of information security attacks.
2. Examine various information security attack vectors.
3. Discuss various information security laws and regulations

REQUIRED READINGS:

Read the following chapters from the book:

- Ethical Hacking Essentials (EHE) v1 Academia Series – Module 01 Information Security Fundamentals

ADDITIONAL RESOURCES:**White Papers:**

- Security Basics: A Whitepaper, <http://networkdls.com/Articles/security101.pdf>

- Introduction to Information Security,
https://www.cengage.com/resource_uploads/downloads/1111138214_259146.pdf
- PCI DSS 101: The Background You Need for Understanding the PCI DSS,
<https://www.newnettechnologies.com/whitepaper/nnt-whitepaper-pci-dss-101.pdf>

Videos:

- Information Security - Basic Concepts, <https://www.youtube.com/watch?v=-rtWd1VfTPE>
- HIPAA Training: What is required for HIPAA Compliance?,
<https://www.youtube.com/watch?v=s9znUYvVO4A>
- What is ISO 27001:2013 by Andi Rafiandi,
<https://www.youtube.com/watch?v=hxbBBI52Qi0>

QUIZ:

1. Which of the following elements of information security ensures that a recipient receives the original message from the sender without any modifications during transit?
 - a. Confidentiality
 - b. Integrity**
 - c. Availability
 - d. Authenticity
2. Which of the following attack vectors involves the use of a huge network of compromised systems by attackers to perform denial-of-service attacks on the target network or systems?
 - a. Botnet**
 - b. Virus
 - c. APT
 - d. Keylogger
3. Which of the following civilian acts enforces “Electronic Transactions and Code Set Standards”?
 - a. HIPAA**
 - b. ISO/IEC
 - c. PCI/DSS
 - d. Sarbanes Oxley Act

WEEK TWO:**MODULES COVERED:**

- Module 02 – Ethical Hacking Fundamentals

WEEK'S OBJECTIVES USED:

2. Comprehend Cyber Kill Chain Methodology, hacking and ethical hacking concepts, hacker classes, different phases of a hacking cycle, and assess essential ethical hacking tools.

WEEK 2 ASSESSMENTS:

These are found in the course under Week Two.

- Discussion Threads: 1 discussion thread (4 pts.)
- Assignments from Readings: 1 Readings (each worth 6 pts.)
- Lab Assignment: 3 lab assignments (each assignment worth 4 pts.)
- Quizzes: 3 quiz questions (each question is worth 1 pt.)

DISCUSSION THREAD:

1. Discuss various phases of hacking cycle.

ASSIGNMENTS FROM READINGS:

1. Explain about the cyber kill chain methodology, attackers' TTPs, and different types of hacker classes/threat actors.

REQUIRED READINGS:

Read the following chapters from the book:

- Ethical Hacking Essentials (EHE) v1 Academia Series – Module 02 Ethical Hacking Fundamentals

ADDITIONAL RESOURCES:**White papers:**

- GAINING THE ADVANTAGE: Applying Cyber Kill Chain® Methodology to Network Defense, https://www.lockheedmartin.com/content/dam/lockheed-martin/rms/documents/cyber/Gaining_the_Advantage_Cyber_Kill_Chain.pdf

- Crackers or Malicious Hackers, <https://www.arsdcollege.ac.in/wp-content/uploads/2020/04/GE-4-1.pdf>
- The 5 Phases Every Hacker Must Follow, https://d3alc7xa4w7z55.cloudfront.net/static/upload/201/0123/2016-ossovernet_ethical_hacking.pdf

Videos:

- What Are Indicators of Compromise (IOC)?, <https://www.youtube.com/watch?v=T9OlnPxpHnc>
- 5 Phases Of Ethical Hacking, <https://www.youtube.com/watch?v=NwMpxUEmE8Y>.
- Top 5 Ethical Hacking Tools | Ethical Hacking Tools And Uses, <https://www.youtube.com/watch?v=Fn4O1rumWrY>

LAB ASSIGNMENTS:**Week 2 Lab Assignment 1: Perform Passive Footprinting to Gather Information About a Target**

Lab Scenario

Passive footprinting involves gathering information about the target without direct interaction. It is mainly useful when the information gathering activities are not to be detected by the target. Performing passive footprinting is technically difficult, as active traffic is not sent to the target organization from a host or anonymous hosts or services over the Internet. We can only collect archived and stored information about the target using search engines, social networking sites, and so on.

Lab Objectives

The objective of this lab is to help students learn and understand how to perform passive footprinting to gather information about a target:

- Gather information using advanced google hacking techniques
- Extract a company's data using Web Data Extractor
- Perform whois lookup using DomainTools

Week 2 Lab Assignment 2: Perform Network Scanning to Identify Live Hosts, Open Ports and Services and Target OS in the Network

Lab Scenario

Network scanning refers to a set of procedures used for identifying hosts, ports, and services in a network. Network scanning is also used for discovering active machines in a network and identifying the OS running on the target machine. It is one of the most important phases of intelligence gathering for an attacker, which enables him/her to create a profile of the target organization. In the process of scanning, the attacker tries to gather information, including the

specific IP addresses that can be accessed over the network, the target's OS and system architecture, and the ports along with their respective services running on each computer.

Lab Objectives

The objective of this lab is to help students learn and understand how to perform network scanning to identify live hosts, open ports and services and target OS in the network:

- Perform network tracerouting in Windows and Linux machines
- Perform host discovery using Nmap
- Perform port and service discovery using MegaPing
- Perform OS discovery using Unicornscan

Week 2 Lab Assignment 3: Perform Enumeration on a System or Network to Extract Usernames, Machine Names, Network Resources, Shares, etc.

Lab Scenario

Enumeration creates an active connection with the system and performs directed queries to gain more information about the target. It extracts lists of computers, usernames, user groups, ports, OSes, machine names, network resources, and services using various techniques. Enumeration techniques are conducted in an intranet environment.

Lab Objectives

The objective of this lab is to help students learn and understand how to perform enumeration on a system or network to extract usernames, machine names, network resources, shares, etc.:

- Perform NetBIOS enumeration using Windows Command-Line utilities
- Perform NetBIOS enumeration using NetBIOS Enumerator

QUIZ:

1. In which of the following phases of the cyber kill chain methodology does the adversary create a tailored malicious payload based on the vulnerabilities identified??
 - a. Installation
 - b. Command and control
 - c. Exploitation
 - d. Weaponization**
2. Which of the following terms refers to the patterns of activities and methods associated with specific threat actors or groups of threat actors that are used to analyze and profile them to enhance an organization's security?
 - a. Tactics, techniques, and procedures**

- b. Tactics, technology, and procedures
 - c. Tactics, tricks, and process
 - d. Tactics, technology, and process
3. Identify the type of hackers recruited by organizations to enhance their cybersecurity by reporting all vulnerabilities to the system and network for remediation.
- a. State sponsor hacker
 - b. Ethical hacker**
 - c. Industrial spy
 - d. Organized hacker

WEEK THREE:**MODULES COVERED:**

- Module 03 – Information Security Threats and Vulnerability Assessment

WEEK'S OBJECTIVES USED:

3. Examine various information security threats and vulnerabilities, identify different types of malware and perform vulnerability assessments.

WEEK 3 ASSESSMENTS:

These are found in the course under Week Three.

- Discussion Threads: 1 discussion thread (4 pts.)
- Assignments from Readings: 1 Readings (each worth 6 pts.)
- Lab Assignment: 3 lab assignments (each assignment worth 4 pts.)
- Quizzes: 3 quiz questions (each question is worth 1 pt.)

DISCUSSION THREAD:

1. Explain about the components of malware and different types of malware.

ASSIGNMENTS FROM READINGS:

1. Explain about vulnerability classification and vulnerability-management life cycle.

REQUIRED READINGS:

Read the following chapters from the book:

- Ethical Hacking Essentials (EHE) v1 Academia Series – Module 03 Information Security Threats and Vulnerability Assessment

ADDITIONAL RESOURCES:**White papers:**

- RANSOMWARE: A look at the criminal art of malicious code, pressure, and manipulation, https://www.welivesecurity.com/wp-content/uploads/2021/08/ransomware_paper.pdf
- Vulnerability Classification, <http://gauss.eecs.uc.edu/Courses/c6056/pdf/vuln-classification.pdf>

- Common Vulnerability Scoring System version 3.1, https://www.first.org/cvss/v3-1/cvss-v31-specification_r1.pdf

Videos:

- Malware | What is Malware | Types of Malware | Computer Virus, Computer Worm, Trojan Horse, Bots, <https://www.youtube.com/watch?v=Lx6MpDCVxi0>
- Protect Your Computer from Malware | Federal Trade Commission, https://www.youtube.com/watch?v=XU8PHihT_P4
- What is Vulnerability assessment and different types of Vulnerability assessment, <https://www.youtube.com/watch?v=5LFcGjGvAZs>

LAB ASSIGNMENTS:**Week 3 Lab Assignment 1: Create a Trojan to Gain Access to the Target System**

Lab Scenario

A Trojan is wrapped within or attached to a legitimate program, meaning that the program may have functionality that is not apparent to the user. Furthermore, attackers use victims as unwitting intermediaries to attack others. They can use a victim's computer to commit illegal DoS attacks.

A compromised system can affect other systems on the network. Systems that transmit authentication credentials such as passwords over shared networks in clear text or a trivially encrypted form are particularly vulnerable. If an intruder compromises a system on such a network, he or she may be able to record usernames and passwords or other sensitive information.

Additionally, a Trojan, depending on the actions it performs, may falsely implicate a remote system as the source of an attack by spoofing, thereby causing the remote system to incur a liability. Trojans enter the system by means such as email attachments, downloads, and instant messages.

The lab tasks in this exercise demonstrate how easily hackers can gain access to the target systems in the organization and create a covert communication channel for transferring sensitive data between the victim computer and the attacker.

Lab Objectives

The objective of this lab is to help students learn and understand how to create a Trojan to gain access to the target system:

- Create a Trojan server using Theef RAT trojan
- Gain control over a victim machine using the njRAT RAT Trojan

Week 3 Lab Assignment 2: Create a Virus to Infect the Target System

Lab Scenario

Viruses are the scourges of modern computing. Computer viruses have the potential to wreak havoc on both business and personal computers. The lifetime of a virus depends on its ability to reproduce. Therefore, attackers design every virus code in such a manner that the virus replicates itself n number of times, where n is a number specified by the attacker.

Virus reproduces its own code while enclosing other executables, and spreads throughout the computer. Viruses can spread the infection by damaging files in a file system. Some viruses reside in the memory and may infect programs through the boot sector. A virus can also be in an encrypted form.

Lab Objectives

The objective of this lab is to help students learn and understand how to create a virus to infect the target system:

- Create a virus using the JPS Virus Maker Tool and infect the target system

Week 3 Lab Assignment 3: Perform Vulnerability Assessment to Identify Security Vulnerabilities in the Target System or Network

Lab Scenario

The information gathered in the previous labs might not be sufficient to reveal potential vulnerabilities of the target: there could be more information available that may help in finding loopholes. As an ethical hacker, you should look for as much information as possible using all available tools. This lab will demonstrate other information that you can extract from the target using various vulnerability assessment tools.

Lab Objectives

The objective of this lab is to help students learn and understand how to perform vulnerability assessment to identify security vulnerabilities in the target system or network:

- Perform vulnerability analysis using OpenVAS

QUIZ:

1. Sam recently joined as a network admin in an organization. He failed to comprehend all the security practices during his training but pretended to have acquired adequate skills. With mediocre knowledge, he left a few loopholes in the firewall implementation that eventually led to unwanted network intrusions.

Identify the threat source that led the way to this perimeter breach.

- a. External threat
- b. Intentional threat

- c. Natural threat
 - d. Unintentional threat**
2. Which of the following malware components hides the malware presence and protects the malware from reverse engineering, thus making it difficult to be detected by security solutions?
- a. Exploit
 - b. Injector
 - c. Crypter**
 - d. Payload
3. Which of the following Trojans can an attacker use for the auto-deletion of files, folders, and registry entries as well as local network drives to cause the operating system to fail?
- a. Defacement Trojan
 - b. Backdoor Trojan
 - c. Destructive Trojan**
 - d. e-Banking Trojan

WEEK FOUR:**MODULES COVERED:**

- Module 04 – Password Cracking Techniques and Countermeasures
- Module 05 – Social Engineering Techniques and Countermeasures

WEEK'S OBJECTIVES USED:

4. Explain and use different password cracking, social engineering, insider threats, and identity theft techniques, and discuss their countermeasures.

WEEK 4 ASSESSMENTS:

These are found in the course under Week Four.

- Discussion Threads: 1 discussion thread (4 pts.)
- Assignments from Readings: 1 Readings (each worth 6 pts.)
- Lab Assignment: 3 lab assignments (each assignment worth 4 pts.)
- Quizzes: 3 quiz questions (each question is worth 1 pt.)

DISCUSSION THREAD:

1. Discuss various type of password attacks.

ASSIGNMENTS FROM READINGS:

1. Explain about various types of social engineering techniques.

REQUIRED READINGS:

Read the following chapters from the book:

- Ethical Hacking Essentials (EHE) v1 Academia Series – Module 04 Password Cracking Techniques and Countermeasures
- Ethical Hacking Essentials (EHE) v1 Academia Series – Module 05 Social Engineering Techniques and Countermeasures

ADDITIONAL RESOURCES:**White papers:**

- Password Cracking: Sam Martin and Mark Tokutomi, <https://www2.cs.arizona.edu/~collberg/Teaching/466-566/2012/Resources/presentations/topic7-final/report.pdf>
- An introduction to social engineering, <https://info.publicintelligence.net/UK-CERT-SocialEngineering.pdf>
- INSIDER THREATS 101: WHAT YOU NEED TO KNOW, https://www.cisa.gov/sites/default/files/publications/Insider%20Threats%20101%20What%20You%20Need%20to%20Know_508.pdf

Videos:

- Password Attacks, https://www.youtube.com/watch?v=aSNzu2-_YnE
- Social engineering Techniques | Types of Social Engineering threats | Social Engineering Hacking, <https://www.youtube.com/watch?v=R45qG4KyWyY>
- What is Identity Theft?, <https://www.youtube.com/watch?v=CygESh4Ki1Q>

LAB ASSIGNMENTS:**Week 4 Lab Assignment 1: Perform Active Online Attack to Crack the System's Password**

Lab Scenario

Active online attack is one of the easiest ways to gain unauthorized administrator-level system access. Here, the attacker communicates with the target machine to gain password access. Techniques used to perform active online attacks include password guessing, dictionary and brute-forcing attacks, hash injection, LLMNR/NBT-NS poisoning, use of Trojans/spyware/keyloggers, internal monologue attacks, Markov-chain attacks, Kerberos password cracking, etc.

The lab in this exercise demonstrates how easily hackers can gather password information from your network and demonstrate the password vulnerabilities that exist in computer networks.

Lab Objectives

The objective of this lab is to help students learn and understand how to perform active online attack to crack the system's password:

- Perform active online attack to crack the system's password using Responder

Week 4 Lab Assignment 2: Audit System Passwords

Lab Scenario

Password auditing is one of the crucial stages in checking the security of a system. Password-auditing mechanisms often exploit otherwise legal means to gain unauthorized system access, such as recovering a user's forgotten password feature. The classification of password attacks depends on the attacker's actions.

The lab in this exercise demonstrates auditing of system passwords using a password auditing tool.

Lab Objectives

The objective of this lab is to help students learn and understand how to audit system passwords:

- Audit system passwords using L0phtCrack
- Audit system passwords using John the Ripper

Week 4 Lab Assignment 3: Perform Social Engineering using Various Techniques to Sniff Users' Credentials

Lab Scenario

In a social engineering test, you should try to trick the user into disclosing personal information such as credit card numbers, bank account details, telephone numbers, or confidential information about their organization or computer system. In the real world, attackers would use these details either to commit fraud or to launch further attacks on the target system.

Lab Objectives

The objective of this lab is to help students learn and understand how to perform social engineering using various techniques to sniff users' credentials:

- Sniff Credentials using the Social-Engineer Toolkit (SET)

QUIZ:

1. Identify the default authentication method in Microsoft operating systems that uses secret-key cryptography for validating identities.
 - a. **Kerberos**
 - b. LM
 - c. NTLM
 - d. SAM
2. Which of the following techniques involves gaining information by listening to someone's conversation or reading private messages being shared between them?

- a. Tailgating
 - b. Dumpster diving
 - c. Eavesdropping**
 - d. Piggybacking
3. Messy, an ex-employee of an organization, was fired because of misuse of resources and security violations. He sought revenge against the company and targeted its network, as he is already aware of its network topology.

Which of the following categories of insiders does Messy belong to?

- a. Malicious insider**
- b. Negligent insider
- c. Professional insider
- d. Compromised insider

WEEK FIVE:**MODULES COVERED:**

- Module 06 – Network Level Attacks and Countermeasures

WEEK'S OBJECTIVES USED:

5. Examine various network-level attacks, including sniffing, denial-of-service, and session hijacking, and discuss their countermeasures.

WEEK 5 ASSESSMENTS:

These are found in the course under Week Five.

- Discussion Threads: 1 discussion thread (4 pts.)
- Assignments from Readings: 1 Readings (each worth 6 pts.)
- Lab Assignment: 3 lab assignments (each assignment worth 4 pts.)
- Quizzes: 3 quiz questions (each question is worth 1 pt.)

DISCUSSION THREAD:

1. Discuss various types of DoS and DDoS attacks.

ASSIGNMENTS FROM READINGS:

1. Explain about various types of sniffing and session hijacking techniques.

REQUIRED READINGS:

Read the following chapters from the book:

- Ethical Hacking Essentials (EHE) v1 Academia Series – Module 06 Network Level Attacks and Countermeasures

ADDITIONAL RESOURCES:**White papers:**

- Sniffer Basics and Detection,
<https://www.just.edu.jo/~tawalbeh/nyit/incs745/presentations/sniffers.pdf>
- Denial of Service (DoS),
<https://www.just.edu.jo/~tawalbeh/nyit/incs745/presentations/DoS.pdf>

- Session Hijacking, <http://index-of.es/Session-Hijacking/Session%20Hijacking.pdf>

Videos:

- Sniffing attack, <https://www.youtube.com/watch?v=U7cmD5z13v0>
- DDoS Attack and Types, <https://www.youtube.com/watch?v=Wan1S9RJacU>
- Software security - Session Hijacking, <https://www.youtube.com/watch?v=fxrCJNQ96Kg>

LAB ASSIGNMENTS:**Week 5 Lab Assignment 1: Perform ARP Poisoning to Divert all Communication Between Two Machines**

Lab Scenario

ARP poisoning technique generally used by attackers to perform sniffing on a target network. Using this method, an attacker can steal sensitive information, prevent network and web access, and perform DoS and MITM attacks using sniffing.

Lab Objectives

The objective of this lab is to help students learn and understand how to perform ARP poisoning to divert all communication between two machines:

- Perform ARP poisoning using arpspoof

Week 5 Lab Assignment 2: Perform DoS and DDoS Attacks using Various Techniques on a Target Host to Prevents Access to System Resources for Legitimate Users

Lab Scenario

DoS and DDoS attacks have become popular, because of the easy accessibility of exploit plans and the negligible amount of brainwork required while executing them. These attacks can be very dangerous, because they can quickly consume the largest hosts on the Internet, rendering them useless. The impact of these attacks includes loss of goodwill, disabled networks, financial loss, and disabled organizations.

In a DDoS attack, many applications pound the target browser or network with fake exterior requests that make the system, network, browser, or site slow, useless, and disabled or unavailable.

The attacker initiates the DDoS attack by sending a command to the zombie agents. These zombie agents send a connection request to a large number of reflector systems with the spoofed IP address of the victim. The reflector systems see these requests as coming from the victim's machine instead of as zombie agents, because of the spoofing of the source IP address. Hence, they send the requested information (response to connection request) to the victim. The victim's machine is flooded with unsolicited responses from several reflector computers at once. This may reduce performance or may even cause the victim's machine to shut down completely.

In this lab, you will gain hands-on experience in auditing network resources against DoS and DDoS attacks.

Lab Objectives

The objective of this lab is to help students learn and understand how to perform DoS and DDoS attacks using various techniques on a target host to prevent access to system resources for legitimate users:

- Perform a DoS attack on a target host using hping3
- Perform a DDoS attack using HOIC

Week 5 Lab Assignment 3: Perform Session Hijacking to Seize Control of a Valid TCP Communication Session Between Two Computers

Lab Scenario

Session hijacking allows an attacker to take over an active session by bypassing the authentication process. It involves stealing or guessing a victim's valid session ID, which the server uses to identify authenticated users, and using it to establish a connection with the server. The server responds to the attacker's requests as though it were communicating with an authenticated user, after which the attacker is able to perform any action on that system.

Attackers can use session hijacking to launch various kinds of attacks such as man-in-the-middle (MITM) and Denial-of-Service (DoS) attacks. A MITM attack occurs when an attacker places himself/herself between the authorized client and the server to intercept information flowing in either direction. A DoS attack happens when attackers sniff sensitive information and use it to make host or network resource unavailable to users, usually by flooding the target with requests until the system is overloaded.

We must possess the required knowledge to hijack sessions in order to test the systems in the target network.

The labs in this exercise demonstrate how to hijack an active session between two endpoints.

Lab Objectives

The objective of this lab is to help students learn and understand how to perform session hijacking to seize control of a valid TCP communication session between two computers:

- Hijack a session using Zed Attack Proxy (ZAP)

QUIZ:

1. Which of the following techniques does an attacker use to compromise the network switches' security for sniffing all the traffic passing through it?
 - a. **MAC flooding**
 - b. UDP flooding

- c. SYN flooding
 - d. DNS spoofing
2. Martin, a hacker, aimed to crash a target system. For this purpose, he spoofed the source IP address with the target's IP address and sent many ICMP ECHO request packets to an IP broadcast network, causing all the hosts to respond to the received ICMP ECHO requests and ultimately crashing the target machine.

Identify the type of attack performed by Martin in the above scenario.

- a. UDP flood attack
 - b. Smurf attack**
 - c. Fragmentation attack
 - d. Multi-vector attack
3. In which of the following phases of session hijacking does an attacker change the sequence number or acknowledgment number (SEQ/ACK) of the server to halt data transmission to a legitimate user?
- a. Command injection
 - b. Session desynchronization**
 - c. Session ID prediction
 - d. Packet injection

WEEK SIX:**MODULES COVERED:**

- Module 07 - Web Application Attacks and Countermeasures

WEEK'S OBJECTIVES USED:

6. Examine various application-level attacks, including webserver exploitation, OWASP top-10 attacks, and SQL injection, and discuss their countermeasures.

WEEK 6 ASSESSMENTS:

These are found in the course under Week Six.

- Discussion Threads: 1 discussion thread (4 pts.)
- Assignments from Readings: 1 Readings (each worth 6 pts.)
- Lab Assignment: 3 lab Assignments (each assignment worth 4 pts.)
- Quizzes: 3 quiz questions (each question is worth 1 pt.)

DISCUSSION THREAD:

1. Discuss various types of web server attacks.

ASSIGNMENTS FROM READINGS:

1. Explain about various types of web application and SQL injection attacks.

REQUIRED READINGS:

Read the following chapters from the book:

- Ethical Hacking Essentials (EHE) v1 Academia Series – Module 07 Web Application Attacks and Countermeasures

ADDITIONAL RESOURCES:**White papers:**

- Web Server Operation,
<http://web.cs.wpi.edu/~kal/courses/awt/lab6/wwwch11servlets.PDF>
- OWASP Top 10 – 2017: The Ten Most Critical Web Application Security Risks,
https://owasp.org/www-pdf-archive/OWASP_Top_10-2017_%28en%29.pdf.pdf

- SQL Injection, <https://www.cisecurity.org/wp-content/uploads/2017/05/SQL-Injection-White-Paper.pdf>

Videos:

- Web Server and Application Server | Explained, <https://www.youtube.com/watch?v=thJSev60yfg>
- Web Application Vulnerability Stack/levels, <https://www.youtube.com/watch?v=AHqsCC6Blbg>
- SQL Injection Attacks - Explained in 5 Minutes, <https://www.youtube.com/watch?v=FHCTfA9cCXs>

LAB ASSIGNMENTS:**Week 6 Lab Assignment 1: Perform a Web Server Attack to Crack FTP Credentials**

Lab Scenario

Attackers perform web server attacks with certain goals in mind. These goals may be technical or non-technical. For example, attackers may breach the security of the web server to steal sensitive information for financial gain, or merely for curiosity's sake. The attacker tries all possible techniques to extract the necessary passwords, including password guessing, dictionary attacks, brute force attacks, hybrid attacks, pre-computed hashes, rule-based attacks, distributed network attacks, and rainbow attacks. The attacker needs patience, as some of these techniques are tedious and time-consuming. The attacker can also use automated tools such as Brutus and THC-Hydra, to crack web passwords.

We must test the company's web server against various attacks and other vulnerabilities. It is important to find various ways to extend the security test by analyzing web servers and employing multiple testing techniques. This will help to predict the effectiveness of additional security measures for strengthening and protecting web servers of the organization.

Lab Objectives

The objective of this lab is to help students learn and understand how to perform a web server attack to crack FTP credentials:

- Crack FTP credentials using a dictionary attack

Week 6 Lab Assignment 2: Perform a Web Application Attack to Compromise the Security of Web Applications to Steal Sensitive Information

Lab Scenario

Attackers perform web application attacks with certain goals in mind. These goals may be either technical or non-technical. For example, attackers may breach the security of the web application and steal sensitive information for financial gain or for curiosity's sake. To hack the web app, first, the attacker analyzes it to determine its vulnerable areas. Next, they attempt to reduce the

“attack surface.” Even if the target web application only has a single vulnerability, attackers will try to compromise its security by launching an appropriate attack. They try various application-level attacks such as injection, XSS, broken authentication, broken access control, security misconfiguration, and insecure deserialization to compromise the security of web applications to commit fraud or steal sensitive information.

We must test their company’s web application against various attacks and other vulnerabilities. They must find various ways to extend the security test and analyze web applications, for which they employ multiple testing techniques. This will help in predicting the effectiveness of additional security measures in strengthening and protecting web applications in the organization.

The task in this lab will assist in performing attacks on web applications using various techniques and tools.

Lab Objectives

The objective of this lab is to help students learn and understand how to perform a web application attack to compromise the security of web applications to steal sensitive information:

- Perform parameter tampering using Burp Suite

Week 6 Lab Assignment 3: Detect SQL Injection Vulnerabilities using SQL Injection Detection Tools

Lab Scenario

By now, you will be familiar with various types of SQL injection attacks and their possible impact. To recap, the different kinds of SQL injection attacks include authentication bypass, information disclosure, compromised data integrity, compromised availability of data and remote code execution (which allows identity spoofing), damage to existing data, and the execution of system-level commands to cause a denial of service from the application.

You must test your organization’s web applications and services against SQL injection and other vulnerabilities, using various approaches and multiple techniques to ensure that your assessments, and the applications and services themselves, are robust.

In this lab, you will learn how to test for SQL injection vulnerabilities using various other SQL injection detection tools.

Lab Objectives

The objective of this lab is to help students learn and understand how to detect SQL injection vulnerabilities using SQL injection detection tools:

- Detect SQL injection vulnerabilities using DSSS

QUIZ:

1. Which of the following web-server components facilitates storage on another machine or disk if the original disk becomes full, in addition to providing object-level security?
 - a. Server root
 - b. Document root
 - c. Virtual document tree**
 - d. Web proxy
2. Which of the following web application vulnerabilities allow untrusted data to be interpreted and executed as part of a command or query?
 - a. Injection flaws**
 - b. Insufficient logging
 - c. Insufficient monitoring
 - d. Broken access control
3. In which of the following attacks does an attacker inject an additional malicious query into an original query to make a DBMS execute multiple SQL queries?
 - a. Piggybacked query**
 - b. Illegal/logically incorrect query
 - c. System stored procedure
 - d. Tautology

WEEK SEVEN:**MODULES COVERED:**

- Module 08 – Wireless Attacks and Countermeasures

WEEK'S OBJECTIVES USED:

7. Discuss different types of wireless encryption, examine wireless threats and attacks, and suggest countermeasures.

WEEK 7 ASSESSMENTS:

These are found in the course under Week Seven.

- Discussion Threads: 1 discussion thread (4 pts.)
- Assignments from Readings: 1 Readings (each worth 6 pts.)
- Lab Assignment: 2 lab Assignments (4 pts.)
- Quizzes: 3 quiz questions (each question is worth 1 pt.)

DISCUSSION THREAD:

1. Discuss different types of wireless network-specific attack techniques.

ASSIGNMENTS FROM READINGS:

1. Discuss various types of wireless encryption and Bluetooth threats.

REQUIRED READINGS:

Read the following chapters from the book:

- Ethical Hacking Essentials (EHE) v1 Academia Series – Module 08 Wireless Attacks and Countermeasures

ADDITIONAL RESOURCES:**White papers:**

- Wireless Networks,
<http://www.ccs.neu.edu/home/rraj/Courses/6710/S10/Lectures/Introduction.pdf>
- KRACK (Key Reinstallation Attack), <https://www.hanwha-security.com/data/tutorial/attrbt/1518156212666.pdf>

- BLUETOOTH HACKING AND ITS PREVENTION,
<https://www.lttts.com/sites/default/files/resources/pdf/whitepapers/2017-12/Bluetooth-Hacking-and-its-Prevention.pdf>

Videos:

- WPA3™: the most advanced Wi-Fi security (English),
https://www.youtube.com/watch?v=gvHhK_LCmr8
- Rogue Access Points, https://www.youtube.com/watch?v=OQLB_RZl2q0
- Bluetooth Attacks (BlueJacking, BlueSnarfing, Blue bugging),
<https://www.youtube.com/watch?v=n1mtr6ckljs>

LAB ASSIGNMENTS:**Week 7 Lab Assignment 1: Perform Wi-Fi Packet Analysis**

Lab Scenario

Our first step in hacking wireless networks is to capture and analyze the traffic of the target wireless network.

This wireless traffic analysis will help you to determine the weaknesses and vulnerable devices in the target network. In the process, you will determine the network's broadcasted SSID, the presence of multiple access points, the possibility of recovering SSIDs, the authentication method used, WLAN encryption algorithms, etc.

The labs in this exercise demonstrate how to use various tools and techniques to capture and analyze the traffic of the target wireless network.

Lab Objectives

The objective of this lab is to help students learn and understand how to perform Wi-Fi packet analysis:

- Wi-Fi packet analysis using Wireshark

Week 7 Lab Assignment 2: Perform Wireless Attacks to Crack Wireless Encryption

Lab Scenario

You must have the required knowledge to perform wireless attacks in order to test the target network's security infrastructure.

After performing the discovery, mapping, and analysis of the target wireless network, you have gathered enough information to launch an attack. You should now carry out various types of attacks on the target network, including Wi-Fi encryption cracking (WEP, WPA, and WPA2), fragmentation, MAC spoofing, DoS, and ARP poisoning attacks.

WEP encryption is used for wireless networks, but it has several exploitable vulnerabilities. When seeking to protect a wireless network, the first step is always to change your SSID from the default before you actually connect the wireless router to the access point. Moreover, if an SSID broadcast is not disabled on an access point, ensure that you do not use a DHCP server, which would automatically assign IP addresses to wireless clients. This is because war-driving tools can easily detect your internal IP address.

You must test its wireless security, exploit WEP flaws, and crack the network's access point keys.

The labs in this exercise demonstrate how to perform wireless attacks using various hacking tools and techniques.

Lab Objectives

The objective of this lab is to help students learn and understand how to perform wireless attacks to crack wireless encryption:

- Crack a WEP network using Aircrack-ng
- Crack a WPA2 network using Aircrack-ng

QUIZ:

1. Which of the following communication standards is also known as WiMAX and is designed to provide multiple physical layer (PHY) and MAC options?
 - a. 802.15.1
 - b. 802.16**
 - c. 802.11n
 - d. 802.11g
2. Which of the following devices does an attacker use to overload traffic to prevent authorized users from accessing a wireless network and block the communication in a certain radius??
 - a. Repeater
 - b. Amplifying device
 - c. Jamming device**
 - d. Routing device
3. Which of the following attacks does an attacker use to breach the security mechanism of a paired Bluetooth device for eavesdropping on the data being carried by short-distance communication protocols?
 - a. Bluejacking
 - b. KNOB attack**
 - c. Bluebugging

d. Bluesmacking

WEEK EIGHT:**MODULES COVERED:**

- Module 09 – Mobile Attacks and Countermeasures

WEEK'S OBJECTIVES USED:

8. Describe various mobile platform attack vectors, mobile device management, mobile security guidelines, and identify essential mobile security tools.

WEEK 8 ASSESSMENTS:

These are found in the course under Week Eight.

- Discussion Threads: 1 discussion thread (4 pts.)
- Assignments from Readings: 2 Readings (each worth 6 pts.)
- Lab Assignment: 2 lab Assignments (4 pts.)
- Quizzes: 3 quiz questions (each question is worth 1 pt.)

DISCUSSION THREAD:

1. Discuss various types mobile platform attacks.

ASSIGNMENTS FROM READINGS:

1. Examine various mobile platform attack vectors, vulnerabilities, and risks.
2. Explain about the mobile device management (MDM) and bring your own device (BYOD) concepts.

REQUIRED READINGS:

Read the following chapters from the book:

- Ethical Hacking Essentials (EHE) v1 Academia Series – Module 09 Mobile Attacks and Countermeasures

ADDITIONAL RESOURCES:**White papers:**

- Security technologies for mobile and BYOD.,
<https://media.kaspersky.com/pdf/kaspersky-mobile-byod-security-technologies-whitepaper.pdf>

- Hacking mobile network via SS7: interception, shadowing and more, <http://secuinside.com/archive/2015/2015-2-7.pdf>
- Mobile Device Management, <https://resources.jamf.com/documents/books/mobile-device-management-101-for-business.pdf>

Videos:

- What exactly is a mobile DEVICE attack?, <https://www.youtube.com/watch?v=noJR25aKDLw>
- What is MDM (Mobile Device Management), https://www.youtube.com/watch?v=ug5KUH_jdA4ent?, https://www.youtube.com/watch?v=OQLB_RZl2q0
- Hacking Mobile Platforms, <https://www.youtube.com/watch?v=GzzSY-m9pHg>

LAB ASSIGNMENT:**Week 8 Lab Assignment 1: Hack an Android Device by Creating Binary Payloads**

Lab Scenario

The number of people using smartphones and tablets is on the rise, as these devices support a wide range of functionalities. Android is the most popular mobile OS, because it is a platform open to all applications. Like other OSes, Android has its vulnerabilities, and not all Android users install patches to keep OS software and apps up to date and secure. This casualness enables attackers to exploit vulnerabilities and launch various types of attacks to steal valuable data stored on the victims' devices.

Owing to the extensive usage and implementation of bring your own device (BYOD) policies in organizations, mobile devices have become a prime target for attacks. Attackers scan these devices for vulnerabilities. These attacks can involve the device and the network layer, the data center, or a combination of these.

You should be familiar with all the hacking tools, exploits, and payloads to perform various tests on mobile devices connected to a network to assess its security infrastructure.

In this lab, we will use various tools and techniques to hack the target mobile device.

Lab Objectives

The objective of this lab is to help students learn and understand how to hack an Android device by creating binary payloads:

- Hack an Android device by creating binary payloads using Parrot Security

Week 8 Lab Assignment 2: Secure Android Devices using Various Android Security Tools

Lab Scenario

Like personal computers, mobile devices store sensitive data and are susceptible to various threats. Therefore, they should be properly secured in order to prevent the compromise or loss of confidential data, lessen the risk of various threats such as viruses and Trojans, and mitigate other forms of abuse. Strict measures and security tools are vital to strengthening the security of these devices.

Android's growing popularity has led to increased security threats, ranging from typical malware to advanced phishing and identity theft techniques. You should scan for any unsecured settings on the mobile device you are assessing, and then take appropriate action to secure them. You must do this before hackers exploit these vulnerabilities by; for example, downloading sensitive data, committing a crime using your Android device as a launchpad, and ultimately endangering your business.

There are various security tools available for scanning, detecting, and assessing the vulnerabilities and security status of Android devices. Many security software companies have launched their own apps, including several complete security suites with antitheft capabilities.

The tasks in this lab will assist you in performing a security assessment of a target Android device.

Lab Objectives

The objective of this lab is to help students learn and understand how to secure Android devices using various Android security tools:

- Secure Android devices from malicious apps using Malwarebytes Security

QUIZ:

1. Which of the following techniques allows attackers to attain privileged control within Android's subsystem, resulting in the exposure of sensitive data?
 - a. OS data caching
 - b. Simjacker
 - c. Carrier-loaded software
 - d. Rooting**
2. Which of the following vulnerabilities leads to erratic program behavior, including memory access errors and incorrect results, and causes mobile devices to crash?
 - a. Clickjacking
 - b. Buffer overflow**
 - c. Data caching
 - d. Phishing

3. Which of the following guidelines must be followed to protect a mobile device from malicious attackers?
- a. Maintain configuration control and management**
 - b. Never install applications from trusted stores
 - c. Always load too many applications and auto-upload photos to social networks
 - d. Always connect to two separate networks, such as Wi-Fi and Bluetooth, simultaneously

WEEK NINE:**MODULES COVERED:**

- Module 10 – IoT and OT Attacks and Countermeasures

WEEK'S OBJECTIVES USED:

9. Discuss IoT and OT concepts, examine various IoT and OT threats and attacks, and suggest countermeasures.

WEEK 9 ASSESSMENTS:

These are found in the course under Week Nine.

- Discussion Threads: 1 discussion thread (4 pts.)
- Assignments from Readings: 2 Readings (each worth 6 pts.)
- Lab Assignment: 2 lab Assignments (4 pts.)
- Quizzes: 3 quiz questions (each question is worth 1 pt.)

DISCUSSION THREAD:

1. Discuss various IoT and OT threats.

ASSIGNMENTS FROM READINGS:

1. Explain various types of IoT attacks and their countermeasures.
2. Explain various types of OT attacks and their countermeasures.

REQUIRED READINGS:

Read the following chapters from the book:

- Ethical Hacking Essentials (EHE) v1 Academia Series – Module 10 IoT and OT Attacks and Countermeasures

ADDITIONAL RESOURCES:**White papers:**

- INTERNET OF THINGS (IOT) SECURITY BEST PRACTICES,
https://internetinitiative.ieee.org/images/files/resources/white_papers/internet_of_things_may_2017.pdf

- Securing the Internet of Things: 5 Key Areas to Protect, <https://www.centritechnology.com/wp-content/uploads/2019/02/CENTRI-Securing-the-IoT-White-Paper-Sept-2016.pdf>
- THE TOP 20 CYBERATTACKS on Industrial Control Systems, <https://www.fireeye.com/content/dam/fireeye-www/products/pdfs/wp-top-20-cyberattacks.pdf>

Videos:

- Challenges Of Internet Of Things, https://www.youtube.com/watch?v=WTSmBr_1MJs
- Armis - BlueBorne Explained, <https://www.youtube.com/watch?v=LLNtZKpLOP8>
- The Top 20 Cyber Attacks on Industrial Control Systems, <https://www.youtube.com/watch?v=dpbJVOh3sb4>

LAB ASSIGNMENTS:**Week 9 Lab Assignment 1: Perform Footprinting using Various Footprinting Techniques**

Lab Scenario

Here, the first step is to gather maximum information about the target IoT and OT devices by performing footprinting through search engines, advanced Google hacking, Whois lookup, etc.

The first step in IoT and OT device hacking is to extract information such as IP address, protocols used (MQTT, ModBus, ZigBee, BLE, 5G, IPv6LoWPAN, etc.), open ports, device type, geolocation of the device, manufacturing number, and manufacturer of the device.

Lab Objectives

The objective of this lab is to help students learn and understand how to perform footprinting using various footprinting techniques:

- Gather information using online footprinting tools

Week 9 Lab Assignment 2: Capture and Analyze IoT Device Traffic

Lab Scenario

You must have sound knowledge to capture and analyze the traffic between IoT devices. Using various tools and techniques, you can capture the valuable data flowing between the IoT devices, analyze it to obtain information on the communication protocol used by the IoT devices, and acquire sensitive information such as credentials, device identification numbers, etc.

Lab Objectives

The objective of this lab is to help students learn and understand how to capture and analyze IoT device traffic:

- Capture and analyze IoT traffic using Wireshark

QUIZ:

1. Which of the following layers of the IoT architecture is responsible for delivering services to respective users from different sectors such as building, industrial, manufacturing, automobile, security, and healthcare?
 - a. Edge technology layer
 - b. Access gateway layer
 - c. Middleware layer
 - d. Application layer**
2. Which of the following layers in the IoT system has security issues such as insecure API, lack of encryption in communication channels, lack of authentication, and lack of storage security?
 - a. Cloud
 - b. Mobile**
 - c. Network
 - d. Application
3. James is a professional hacker attempting to gain access to an industrial system through a remote control device. In this process, he used a specially designed radio transceiver device to sniff radio commands and inject arbitrary code into the firmware of the remote controllers to maintain persistence.

Which of the following attacks is performed by James in the above scenario?

 - a. Re-pairing with a malicious RF controller
 - b. Malicious reprogramming attack**
 - c. Abusing reprogramming attack
 - d. Command injection

WEEK TEN:**MODULES COVERED:**

- Module 11 – Cloud Computing Threats and Countermeasures
- Module 12 – Penetration Testing Fundamentals

WEEK'S OBJECTIVES USED:

10. Describe various Cloud computing technologies, examine cloud computing threats and attacks, and identify security techniques.
11. Discuss fundamentals of penetration testing, its benefits, strategies, and phases, and examine guidelines for penetration testing.

WEEK 10 ASSESSMENTS:

These are found in the course under Week Ten.

- Discussion Threads: 1 discussion thread (4 pts.)
- Assignments from Readings: 1 Readings (each worth 6 pts.)
- Lab Assignment: 2 lab Assignments (4 pts.)
- Quizzes: 7 quiz questions (each question is worth 1 pt.)

DISCUSSION THREAD:

1. Discuss various types of cloud attacks and various strategies of penetration testing.

ASSIGNMENTS FROM READINGS:

1. Explain about cloud deployment models and various phases of penetration testing.

REQUIRED READINGS:

Read the following chapters from the book:

- Ethical Hacking Essentials (EHE) v1 Academia Series – Module 11 Cloud Computing Threats and Countermeasures
- Ethical Hacking Essentials (EHE) v1 Academia Series – Module 12 Penetration Testing Fundamentals

ADDITIONAL RESOURCES:**White papers:**

- Introduction to Cloud Computing,
<https://www.kth.se/social/files/554fa451f276544829be2e5e/9-cloud-computing.pdf>
- Cloud Deployment Models and Cloud Services Models,
<https://www.dauniv.ac.in/public/frontassets/coursematerial/InternetofThings/IoTCh06L03CloudDeployment&ServiceModels.pdf>
- Vulnerability Assessment & Penetration Testing,
<https://www.secura.com/uploads/factsheets/Vulnerability-Assessment-Penetration-Testing.pdf>

Videos:

- The NIST Cloud Computing Reference Architecture,
<https://www.youtube.com/watch?v=nRdNgMcKge8>
- Virtual Machines vs Containers - Which is right for you?,
<https://www.youtube.com/watch?v=XCWWPpfdbmM>
- MicroNugget: What is Black vs Gray vs White Box Testing?,
<https://www.youtube.com/watch?v=C5fUkYMV1-0>

LAB ASSIGNMENTS:**Week 10 Lab Assignment 1: Perform S3 Bucket Enumeration using Various S3 Bucket Enumeration Tools**

Lab Scenario

Enumeration tools are used to collect detailed information about target systems to exploit them. Information collected by S3 enumeration tools consists of a list of misconfigured S3 buckets that are available publicly. Attackers can exploit these buckets to gain unauthorized access to them. Moreover, they can modify, delete, and exfiltrate the bucket content.

In this lab, you must try to obtain as much information as possible about the target cloud environment using various enumeration tools. This lab will demonstrate various S3 bucket enumeration tools that can help you in extracting the list of publicly available S3 buckets.

Lab Objectives

The objective of this lab is to help students learn and understand how to perform S3 bucket enumeration using various S3 bucket enumeration tools:

- Enumerate S3 buckets using lazys3

Week 10 Lab Assignment 2: Exploit S3 Buckets

Lab Scenario

S3 buckets are used by customers and end users to store text documents, PDFs, videos, images, etc. To store all these data, the user needs to create a bucket with a unique name.

In this lab, you must have sound knowledge of enumerating S3 buckets. Using various techniques, you can exploit misconfigurations in bucket implementation and breach the security mechanism to compromise data privacy. Leaving the S3 bucket session running enables you to modify files such as JavaScript or related code and inject malware into the bucket files. Furthermore, finding the bucket's location and name will help you in testing its security and identifying vulnerabilities in the implementation.

Lab Objectives

The objective of this lab is to help students learn and understand how to exploit S3 buckets:

- Exploit open S3 buckets using AWS CLI

QUIZ:

1. Which of the following types of cloud services provides data processing services, such as IoT services for connected devices, mobile and web applications, and batch-and-stream processing?
 - a. **Function-as-a-service (FaaS)**
 - b. Identity-as-a-service (IDaaS)
 - c. Security-as-a-service (SECaaS)
 - d. Platform-as-a-service (PaaS)
2. Identify the node component of Kubernetes that ensures all pods and containers are healthy and running as expected.
 - a. Kube-apiserver
 - b. Kube-proxy
 - c. **Kubelet**
 - d. Container runtime
3. In which of the following attacks does an attacker exploit the vulnerability in a bare-metal cloud server and use it to implant a malicious backdoor in its firmware?
 - a. Cloud hopper attack
 - b. **Cloudborne attack**
 - c. Man-in-the-cloud

- d. Cloud Cryptojacking
- 4. Which of the following best practices should be followed for securing a cloud environment?
 - a. Do not disclose applicable logs and data to customers
 - b. Never apply a baseline security breach notification process
 - c. Do not enforce legal contracts in employee behavior policy
 - d. Verify one's own cloud in public domain blacklists**
- 5. Which of the following activities is implemented to check whether an organization is following a set of standard policies and procedures in protecting its network?
 - a. Ethical hacking
 - b. Vulnerability assessment
 - c. Security audit**
 - d. Penetration testing
- 6. Steve, a professional pen tester, was hired by an organization to assess its cybersecurity. The organization provided Steve with details such as network topology documents, asset inventory, and valuation information. This information helped Steve complete the penetration test successfully, and he provided a snapshot of the organization's current security posture.

Identify the penetration testing strategy followed by Steve in the above scenario.

 - a. Black-box testing
 - b. White-box testing**
 - c. Grey-box testing
 - d. Goal-oriented penetration testing
- 7. Which of the following guidelines helps a penetration tester minimize risks and avoid DoS conditions while performing penetration testing?
 - a. Use direct testing
 - b. Perform uninterruptible testing
 - c. Use reserved addresses**
 - d. Directly exploit vulnerabilities

WEEK ELEVEN:**WEEK'S OBJECTIVES USED:**

1. Discuss the key issues plaguing the information security world and review various information security laws and regulations.
2. Comprehend Cyber Kill Chain Methodology, hacking and ethical hacking concepts, hacker classes, different phases of a hacking cycle, and assess essential ethical hacking tools.
3. Examine various information security threats and vulnerabilities, identify different types of malware and perform vulnerability assessments.
4. Explain and use different password cracking, social engineering, insider threats, and identity theft techniques, and discuss their countermeasures.
5. Examine various network-level attacks, including sniffing, denial-of-service, and session hijacking, and discuss their countermeasures.
6. Examine various application-level attacks, including webserver exploitation, OWASP top-10 attacks, and SQL injection, and discuss their countermeasures.
7. Discuss different types of wireless encryption, examine wireless threats and attacks, and suggest countermeasures.
8. Describe various mobile platform attack vectors, mobile device management, mobile security guidelines, and identify essential mobile security tools.
9. Discuss IoT and OT concepts, examine various IoT and OT threats and attacks, and suggest countermeasures.
10. Describe various Cloud computing technologies, examine cloud computing threats and attacks, and identify security techniques.
11. Discuss fundamentals of penetration testing, its benefits, strategies, and phases, and examine guidelines for penetration testing.

WEEK 11 ASSESSMENTS:

Summative Assessment Research Project: 50 pts.

Directions on Project:

Your instructor will provide information about project topics, scope of work, and submission requirements. Review scoring rubric for project to aid in the construction and organization of the project. Listed below are potential guidelines your instructor **MAY** follow.

Guidelines on Graduate Project

Following are the guidelines for a graduate project. Apart from these guidelines, your instructor may also suggest guidelines and instructions as required during the course of the graduate project.

Selecting a Topic

- Choose a project on a topic of your interest related to your subject of study. The idea for a project may come from your own curiosity, from your coursework, interactions with colleagues, faculty members, and general observation.
- Explore the idea; examine its significance and feasibility. Go through the existing projects on similar topics. Conduct a preliminary research on the idea. Review the relevant literature. Examine the gap areas in research and identify issues you want to address. The idea should be new or must add significant value to the existing projects.
- Prepare a rough outline of project proposal you would like to submit. Discuss the topic and draft project proposal with your project coordinator, exchange ideas and incorporate suggestions.
- Conduct further research on the topic. Make a detailed proposal. In the project proposal you should:
 - a. Introduce the topic
 - b. Explain your rationale for selecting the project
 - c. Describe significance of the project
 - d. State the objective of the project and project outline
 - e. Describe the methodology to be adopted
 - f. State the timeline for the project completion
 - g. Include references
- Finalize the proposal with your instructor by week five (specific instructors may vary that date)

Working on the Project

- On approval of the project, you should start working on your project. It is recommended you do this prior to the 5th week of the course, but speak with your instructor to clarify dates. Limit your research to the approved proposal.
- You have to complete your project within the stipulated deadlines. Plan your project accordingly
- While meeting the executives of a company in relation to your project, make sure you have appropriate approvals and request letters from the concerned university department.
- Make sure your instructor approves questionnaires designed for any survey in relation to the project.
- You must use any data collected in course of the research, only for the approved project. You must not share collected information with other candidates.

- Make notes of key points during the course of research. It would save lot of time in preparation of project report.
- Make sure all relevant journals, magazines, papers and books are available in the university library. Please check with the library information desk for information on resources, currently not available with the library.
- Analysis is the most critical part of the project and forms basis for all findings. Make sure you make use of appropriate statistical tools in analysis. Take guidance of your project coordinator during the analysis. Do validate your findings with your project coordinator.

Writing a Project Report

- Review the style guidelines for project report
- The project report should not exceed 7,000 words
- Abstract should be between 150-250 words
- Select A4 size; page orientation should be portrait. Specify “1” margin on all sides.
- Number all pages consecutively. Start every chapter on a new page.
- Provide double spacing
- You should use Times New Roman Font- “12” for text and “10” for footnotes. Use a larger font size for section headings.
- A project report must contain:

Content	Section
a. Title Page	Preliminaries
b. Table of Contents	
c. Abstract	
d. Introduction and background	Body of the report
e. Problem statement	
f. Objectives of the project	
g. Literature review	
h. Methodology adopted	
i. Results - project findings	
j. Recommendations	
k. Conclusion	
l. Bibliography	References
m. Appendix	
n. List of figures and tables	

o. Index words (if required)	
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- Be clear and precise. Express your ideas in a logical way.
- Abstract should reflect the essence of the project
- The introduction should provide the overview of the topic and highlight its significance
- Clearly indicate the objectives of your project.
- Describe all the methods used such as interviews, questionnaires in the methodology section.
- Ensure that literature review is in your own words. Analyze other person's contribution to the topic. Identify the gaps in the literature. Emphasize on the likely contribution of your project to the existing literature on the topic.
- Describe your findings from analysis in the results section. As this is the most critical part of the project, ensure that there are no errors in analysis. Make proper inferences from analysis and findings.
- The conclusion section should summarize your objectives, findings and learning's from the project. Provide useful supplementary information in the Appendix.
- Avoid plagiarism. The project report should reflect your understanding of the topic. The majority of the paper should be in your own words and reflect your own ideas.
- Give credit for all referenced work. Provide appropriate citation and references for all quotations.
- Ensure that papers referenced are relevant and not outdated.
- Your paper should be reader friendly. Use footnotes to explain difficult terms.
- Don't use text from Wikipedia in footnotes
- All tables and figures must be suitably numbered and titled. Give appropriate credit.
- On completion, go through the entire project. Ensure there are no proofing errors and you have adhered to all guidelines related to the project.
- Submit spiral bound copies of the project to your project coordinator and the project review committee. If suggested, create a soft copy of the project in PDF format and submit it relevant authorities.

WEEK TWELVE:**WEEK'S OBJECTIVES USED:**

1. Discuss the key issues plaguing the information security world and review various information security laws and regulations.
2. Comprehend Cyber Kill Chain Methodology, hacking and ethical hacking concepts, hacker classes, different phases of a hacking cycle, and assess essential ethical hacking tools.
3. Examine various information security threats and vulnerabilities, identify different types of malware and perform vulnerability assessments.
4. Explain and use different password cracking, social engineering, insider threats, and identity theft techniques, and discuss their countermeasures.
5. Examine various network-level attacks, including sniffing, denial-of-service, and session hijacking, and discuss their countermeasures.
6. Examine various application-level attacks, including webserver exploitation, OWASP top-10 attacks, and SQL injection, and discuss their countermeasures.
7. Discuss different types of wireless encryption, examine wireless threats and attacks, and suggest countermeasures.
8. Describe various mobile platform attack vectors, mobile device management, mobile security guidelines, and identify essential mobile security tools.
9. Discuss IoT and OT concepts, examine various IoT and OT threats and attacks, and suggest countermeasures.
10. Describe various Cloud computing technologies, examine cloud computing threats and attacks, and identify security techniques.
11. Discuss fundamentals of penetration testing, its benefits, strategies, and phases, and examine guidelines for penetration testing.

WEEK 12 ASSESSMENTS:**Summative Assessment Final Exam:** 100 pts.

During week twelve you will be required to complete a final exam that incorporates several areas of critical thinking and decision-making. The exam will include multiple choice, short answer, and essay questions covering all learning outcomes of the course. Contact your instructor for specific question related to this assessment.